

- VICTOR BARROSO-NASCIMENTO, *Higher-order Kripke Models for Intuitionistic and Other Non-classical Modal Logics*.

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This talk introduces higher-order (“nested”) Kripke models, a generalization of standard Kripke models that is remarkably close to Kripke’s original idea – both mathematically and conceptually. Standard Kripke models are now considered 0-ary models, whereas n -ary models for $n > 0$ are models whose set of objects (“possible worlds”) contains only $(n - 1)$ -ary Kripke models. This framework is obtained by promoting a radical change of perspective in how modal semantics for non-classical logics are defined: just like classical modalities are obtained through accessibility relations between classical propositional models, non-classical modalities are now obtained through accessibility relations between non-classical propositional models (even when they are Kripke models). The talk introduces the models after dealing specifically with intuitionistic modal logic. It is shown that, depending on which intuitionistic 0-ary propositional models are allowed, we obtain 1-ary models equivalent to either birelational models for IK or for a new logic called MK . Extensions to WK and CK are briefly discussed, as are models with infinitely many layers. Those 1-ary models have an intuitive reading that adds to the interpretation of intuitionistic models in terms of “timelines” the concept of “alternative timelines”. More generally, n -ary models (for $n > 0$) can be read as defining a concept of “alternative” for any interpretation of the $(n - 1)$ -ary models. The semantic clauses for necessity and possibility of MK are also modular and can be used to obtain similar modal semantics for every non-classical logic, each of which can be provided with a similar intuitive reading.